Mohammad Shaharyar Ahsan

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I am pursuing a Computer Science degree at LUMS, and am passionate about data science, machine learning and NLP, particularly in the fields of computational linguistics and musicology, and NLP for social impact.

Education

Lahore University of Management Sciences (LUMS) — B.S. Computer Science Sept 2021 - June 2025

- **GPA:** 3.80/4.0 **, Major GPA:** 3.87/4.0
- **Relevant Coursework:** Machine Learning, Intro to Artificial Intelligence, Principles and Techniques of Data Science, Probability, Topics in Large Language Models, Internet of Things, Generative AI, Tech and Text: Digital Approaches to the Humanities, Databases, Internet Governance and Technology Policy

Awards & Honors

- **Dean's Honor List and Award of High Distinction**, Academic Year 2024-2025
- **Dean's Honor List**, Academic Year 2023-2024
- **Dean's Honor List**, Academic Year 2022-2023
- **Merit Scholarship** - O and A Levels, Academic Year 2021-2022 (LUMS)
- **Outstanding Cambridge Learner Award** in O Level English Literature, 2019

Research Experience

Research Assistant — Embedded AI Lab, LUMS June 2024 - July 2025

Advisors: Dr Naveed Anwar Bhatti (Assistant Professor), Dr Hamad Alizai (Associate Professor)

Worked on the Guardian Angel project, designing a low-cost, screenless wearable paired with a WhatsApp-based LLM agent to democratize health monitoring beyond economic and linguistic obstacles

- Implemented a small-scale prototype initially; Rebuilt the project in the Embedded AI lab, upgrading the hardware to use an ATmega328P MCU and off-the-shelf sensors (MAX30102, ADXL345) to minimize cost and maximize battery life.
- Used the WhatsApp Business API and a tiered LLM backend (GPT-4o-mini, o3-mini, o1) to build the chatbot, creating a seamless interface for users with low digital literacy.
- Tracked the project budget and led a team of three, coordinating weekly with research supervisors to meet deployment deadlines.
- Helped write C/C++ firmware and determine the optimal sensor suite for the final PCB version, implementing duty-cycled sampling to extend battery longevity.
- Learned about GATT Bluetooth protocols and used them to transmit data using an HM-10 BLE module, implementing a custom packet structure to prevent buffer overflows.
- Sourced light libraries for use on a space-constrained MCU, notably fixing critical signal noise issues by implementing on-device filtering for the accelerometer and PPG sensors.
- Sourced a BLE library on GitHub (FastBLE) and wrote an Android app in Java for secure data bridging, intermediate processing (JSON serialization), and background transmission to the inference server.
- Wrote and deployed an inference server in Node.js, engineering a "sequence-to-values" interpreter that uses LLMs to estimate physiological data from raw noisy waveforms.
- Built function-calling capabilities into the server by adding tools for RAG (Medical Literature), charting, and cron-job scheduling for reminders.
- Created a vector-store in Pinecone, sourcing medical documents and setting up a Python pipeline for vectorizing and storing them to ground the LLM's advice in clinical guidelines.
- Conducted extensive testing to validate system robustness, achieving 100% data availability (vs. 70.29% for standard algorithms) and reducing Heart Rate Mean Absolute Error by 46%.
- Implemented a model routing system to amortize costs based on query complexity, achieving a 56.57% relative reduction in inference costs.
- Tested the device in a 96-hour micro-longitudinal study with 20 participants, conducting semi-structured interviews and thematic analysis which revealed significant improvements in health data comprehension and mindfulness of vital signs.

[1] sherryportfolio.vercel.app
[2] linkedin.com/in/shaharyar-ahsan13/
[3] github.com/swiftiecoder

Research Intern — Center for Speech & Language Technologies, LUMS Jan 2024 - May 2024

Advisors: Dr Agha Ali Raza (Assistant Professor)

Worked on the Khpal Tabib project, backed by funding from the Gates Foundation. The project involved creating an AI chatbot for Pashto speakers to allow them to have accessible healthcare where the penetration of clinics and healthcare professionals is low in their native language

- ★ Conducted a thorough literature review on biomedical NLP research, identifying articles which aligned with the research goals of the project and outlining how they did so. Focused on biomedical prompt engineering and alignment.
- ★ Wrote a python script which allowed me to use a Reddit API wrapper to scrape the moderated AskDocs subreddit for responses by medical professionals and create a dataset with the alpaca prompt format, using regex methods for cleaning and preprocessing the text.
- ★ Trained a Gemma model using a pooled dataset with Unsloth, saving the LoRa adapters and model weights for reproducibility.
- ★ Suggested possible evaluation metrics sourced from state-of-the-art industry methods and the latest research.

Employment History

Developer Advocate — Educative June 2025 - Present

- ★ Engineered custom automation tools — including Python-based data scrapers and a proprietary Chrome Extension — to track engagement and optimize workflows, saving ~20 monthly hours.
- ★ Orchestrated a strategic reputation turnaround, utilizing customer sentiment analysis to elevate the Trustpilot score from 2.5 to 4.1 stars and leveraging those insights to guide the Product team's development roadmap.
- ★ Architected an executive content pipeline, using an Opal automation to source industry trends and ghostwrite technical thought leadership for the CEO, significantly amplifying page impressions and engagement.
- ★ Produced high-fidelity technical demos (Cloud Labs, System Design) and led product marketing efforts surrounding social and email campaigns — including running and promoting two successful giveaway challenges.
- ★ Revamped the developer engagement strategy by shifting to persona-based technical content, driving a 26.3% increase in LinkedIn page views and fostering cross-functional alignment between Engineering and Marketing.

Teaching Assistant — Suleman Dawood School of Business, LUMS Sept 2024 - Dec 2024

Course: MGMT 348 – Internet Governance and Technology Policy

- ★ Marked qualitative CP in weekly sessions (2x weekly), as well as offered feedback on how to improve for struggling students.
- ★ Helped create, invigilate and check quizzes, exams and in-class writing assignments over the course of the semester, offering students constructive criticism in contestations. This included limiting the use of AI by negatively marking and offering feedback.
- ★ Helped manage the logistics of presentations, such as scheduling conflicts and auditorium bookings for block classes.
- ★ Held office hours regularly to offer struggling students feedback and help them improve.
- ★ Assisted the professor by using a database join system to speed up data entry on the LUMS portal for the final grading.

Teaching Assistant — Syed Babar Ali School of Science and Engineering , LUMS Sept 2023 - Dec 2023

Course: CS 100 – Computational Problem Solving

- ★ Assisted in creating weekly lab tasks which tested concepts taught in the lectures for that week, as well as the marking schemes for that lab.
- ★ Invigilated labs to provide feedback to struggling students, as well as checked the lab tasks either in-lab with feedback or after the fact. Held contestations to allow students to discuss their issues and understand mistakes.
- ★ Invigilated graded components such as the midterm, final and quizzes held throughout the semester.
- ★ Was the lead for a subset of students for their final project. Helped them finalize their ideas, provided guidance for design and implementation and graded their projects.
- ★ Held office hours weekly to provide struggling students with feedback.

Teaching Assistant — Mushtaq Ahmad Gurmani School of Humanities and Social Sciences, LUMS Sept 2023 - Dec 2023

Course: SS 100 – Writing and Communication

- ★ Marked attendance and class participation in weekly sessions (2x lectures a week).
- ★ Assisted in grading components of the course, such as presentations, quizzes and writing assignments.
- ★ Held office hours weekly to provide struggling students with feedback and allow contestations.
- ★ Taught three sessions on conducting academic research and formatting citations, as well as separate seminars for individualized feedback to help students narrow their chosen research question for the final essay.

Academic Projects

Hooked on a Feeling - A Computational Analysis of Songwriting as Cultural Memory Sept 2024 - Dec 2024

Course: Tech and Text: Digital Approaches to the Humanities

★ Used data science and machine learning methods to conduct a sentiment, linguistic, and musicological analysis of 353 Grammy Song of the Year nominees from 1960 to 2025.
★ Built a multi-stage scraping pipeline using Python (BeautifulSoup, Dask) to compile a dataset of nominees, using LyricGenius for text and the Spotify Web API for audio features.
★ Used DistilBERT (fine-tuned on SST-2) for sentiment tracking and TF-IDF for thematic modeling to quantify shifts in lyrical content over seven decades.
★ Made arguments to support my thesis regarding an affective inversion — identifying that lyrical optimism peaked in the 1980s and shifted to sustained pessimism in the 2020s — supported by qualitative close reading of case studies.
★ Identified and quantified "multimodal dissonance" in modern pop, where sad lyrics are increasingly packaged in high-energy, electronic soundscapes.
★ Accepted for publication at the journal Digital Studies / Le champ numérique (DSCN)/

Portal-LLM: Character-based Vicarious Learning through Chatbots Sept 2024 - Dec 2024

Course: Topics in LLMs: Systems, Applications & Impacts

★ Identified a lack of interest in traditional reading practices in the education vertical and used existing literature to identify why students are uninterested in long-form narratives.
★ Designed and built a project which aimed to transfer the useful aspects of vicarious learning through the experiences of characters in narrative form to text-messaging using LLMs to appeal to younger audiences using the Telegram API.
★ Built a RAG pipeline which automatically vectorized documents sent in the chat, using Pinecone as the vector-store, as well as a web retriever which added web results.
★ Built an LLM QA pipeline which classified prompts into ‘code-switching’ to allow changing characters in the chat and ‘question-answering’, which drew from the vector store and used an engineered prompt to answer as those characters.
★ Tested the performance of multiple models from the Google family, including Gemini 1.5 Flash and 2.0 Experimental, as well as LearnLM 1.5, a fine-tuned model for educational tasks.
★ Tested the performance of two embedding styles offered by Google to see which would perform best for our use-case.
★ Tested the accuracy of the results using a Harry Potter trivia dataset and manually annotating results, achieving an accuracy of 83% in the best case.

Tabular Regression on the Blue Book for Bulldozers dataset Sept 2023 - Dec 2023

Course: Machine Learning

★ Used the Blue Book for Bulldozers dataset on Kaggle to perform 5 experiments using various ML models
★ Extensively pre-processed the data, exploring mechanisms for filling in null data, standardizing representations, using encoding mechanisms to represent categorical data etc.
★ Compared multiple methods for feature extraction, such as Mutual Information Gain and correlation matrices
★ Trained and compared multiple models, including Multiple Regressions, K-Neighbors regressor, Neural Network, SVM and Random Forest. Used standard metrics for error such as R^2 and RMSE.

Heart Disease Classification Jan 2023 - May 2023

Course: Principles and Techniques of Data Science

★ Used a heart disease dataset on Kaggle to conduct a data science project related to EDA and model training
★ Cleaned the dataset by replacing or dropping missing data and generated encodings for categorical labels
★ Used box-and-whisker plots and histograms to analyze the data before and after cleaning to iteratively judge the spread of data and the effectiveness of the cleaning process
★ Plotted correlations to identify useful features using density plots, verifying our choice of features by using bootstrap sampling
★ Trained multiple models (SVM, Logistic Regression, KNN, Decision Tree, Gradient Boosting and CatBoost) and plotted the confusion matrix for each to see which would perform best, achieving 84%+ accuracy for all, with the best one being 90%.
★ Wrote and published a blog post detailing the process on Medium.

On Geoblocking Sept 2023 - Dec 2023

Course: Internet Governance and Technology Policy

★ Acting as a fictional NGO, our group created hypothetical policy guidelines to prevent geo-blocking as a mechanism for power retention for our final presentation.

- ★ Conducted extensive research on how geo-blocking manifests in third-world countries, particularly in terms of healthcare and content creation, and related existing policy measures.
- ★ Suggested policy measures which can be used to allow greater distribution of useful features globally, advocating for a more open and inclusive internet by citing how walled apps for diabetes monitoring, for instance, can be harmful
- ★ Included a budgeting plan for implementing suggesting policies
- ★ Conducted and presented a short case-study on Spotify, exploring how geo-blocked content and features can have implications for the market

Zaamin: A Secure and Compliant Data Portal Jan 2024 - May 2024

Course: Software Engineering

- ★ Designed and hosted an HR portal using Agile methodology throughout the project, externally evaluated by a software house (Devsinc). Alternated as Scrum master, frontend/backend developer, and designer.
- ★ Maintained extensive project documentation, including an SRS, SDS, and a GitHub project board
- ★ Designed the application on Figma, implemented it using React JS, HTML and CSS and deployed both the front-end and back-end on DigitalOcean and Vercel.
- ★ Implemented optimizations such as robust encryption methods (AES-256 and AES-CBC) for data protection.
- ★ Implemented audit logging and role-based access control (RBAC), as well as two-factor authentication using external APIs like the Hunter API, Visual Crossing and EmailJS.

Publications & Working Papers

- ★ **Hooked On A Feeling: A Computational Approach Towards Understanding Songwriting As Cultural Memory -** Published by Digital Studies / Le champ numérique (DSCN), https://doi.org/10.16995/dscn.23539
- ★ **LLM-Enhanced Wearables for Comprehensible Health Guidance in LMICs -** Manuscript Submitted to and Peer-Reviewed at EWSN, IMWUT, PerCom, Sensys; Pre-print available at https://doi.org/10.48550/arXiv.2602.08701
- ★ **Poster: Bridging IoT Gaps in Developing Regions with LLMs -** Poster at EWSN '24, https://ewsn.org/file-repository/ewsn2024/ewsn2024posters-final3.pdf

Skills & Toolkit

- ★ **Research Methods:** Qualitative and Quantitative Research, Literature Reviews, Dataset Generation, System Design
- ★ **Data Analysis Tools:** SQL, MongoDB, Pandas, Matplotlib, NLTK, Vader, Sci-Kit Learn, HuggingFace, MATLAB
- ★ **Design and Prototyping:** Figma, LucidChart
- ★ **Proficient Languages:** Kotlin, MERN Stack, Python, R, Java, JavaScript and TypeScript, C, C++, HTML, CSS, LaTeX, Strudel, Haskell, Bash, Markdown
- ★ **Project Management:** GitHub, Git (version control), Agile methodologies, Monday.com
- ★ **Soft Skills:** Presentations, Leadership, Teaching, Team management

Certifications

- ★ **Learn R - Educative**
- ★ **MCP Fundamentals for Building AI Agents — Educative**
- ★ **Getting Started with AWS Key Management Service (KMS) — Educative**
- ★ **Building Multi-Agentic AI Workflows Using Amazon Bedrock — Educative**

Leadership & Extracurriculars

Member - Feminism Society (FemSoc) at LUMS Sept 2021 - May 2022

- ★ Helped organize and promote social media marketing and physical campaigns
- ★ Advertised, helped organize and attended panel talks on pertinent issues
- ★ Actively participated in reading circles to learn and promote feminist ideas and literature.

Member - Drama Society (Dramaline) at LUMS Sept 2021 - May 2022

- ★ Collaborated with the team to design and caption social media posts to attract a larger audience
- ★ Helped organize the 'Skit Tamasha' event by calling and networking with potential sponsors, as well as promoting the event on social media
- ★ Helped organize stalls to promote the annual play and film promotional material at them.